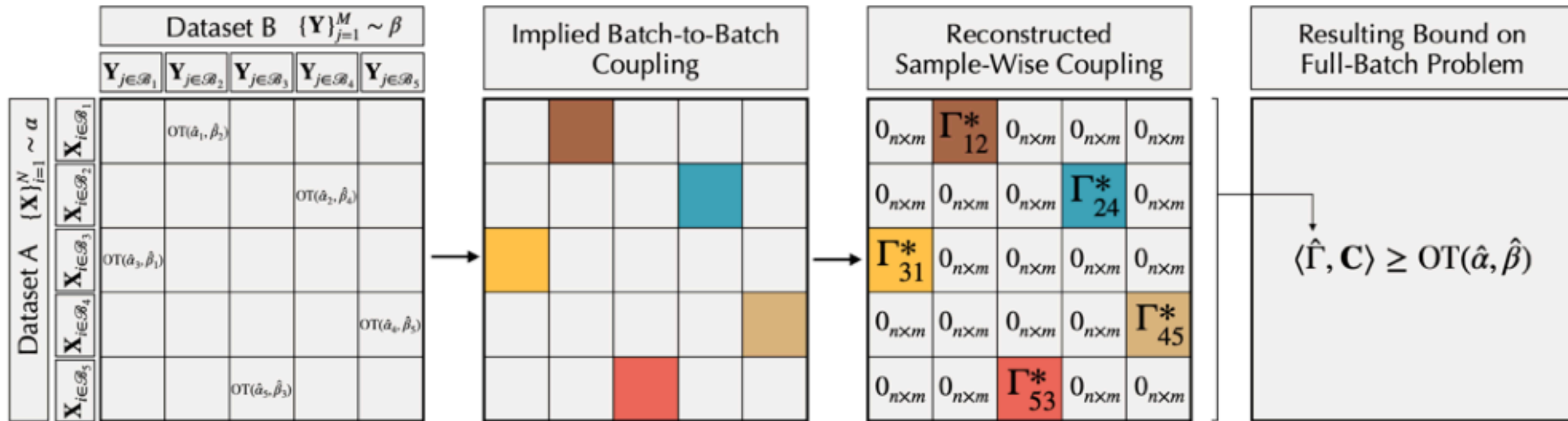
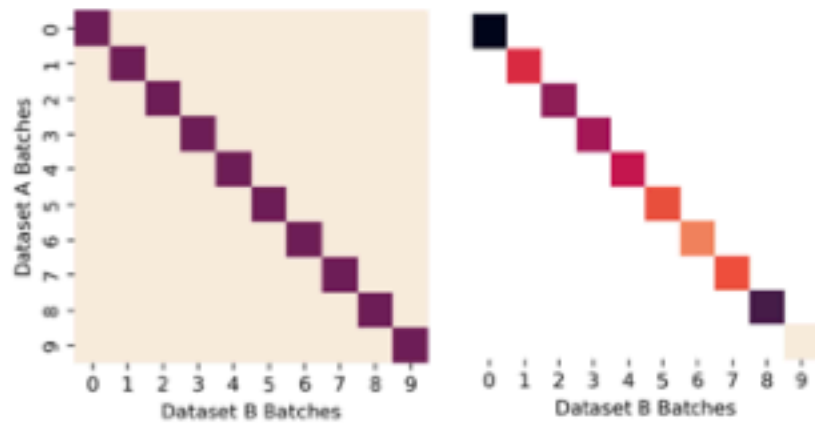


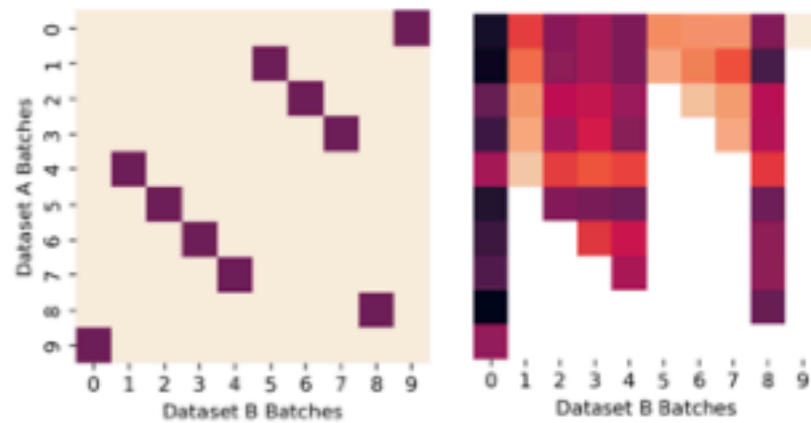
useful gradients

it's an upper bound

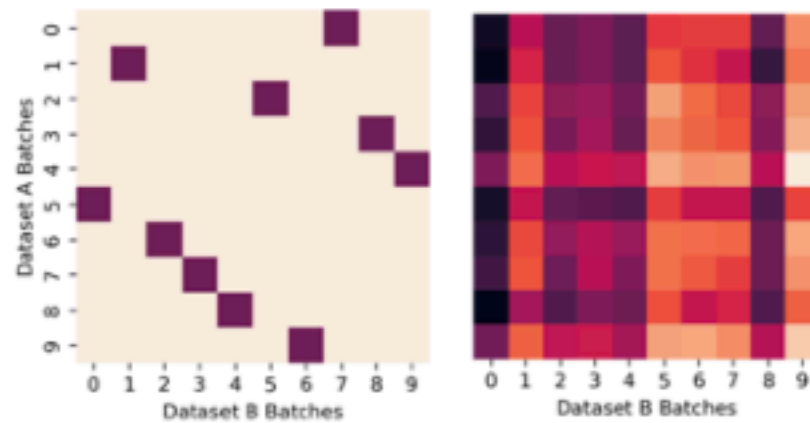




(a) Naive random batch matching



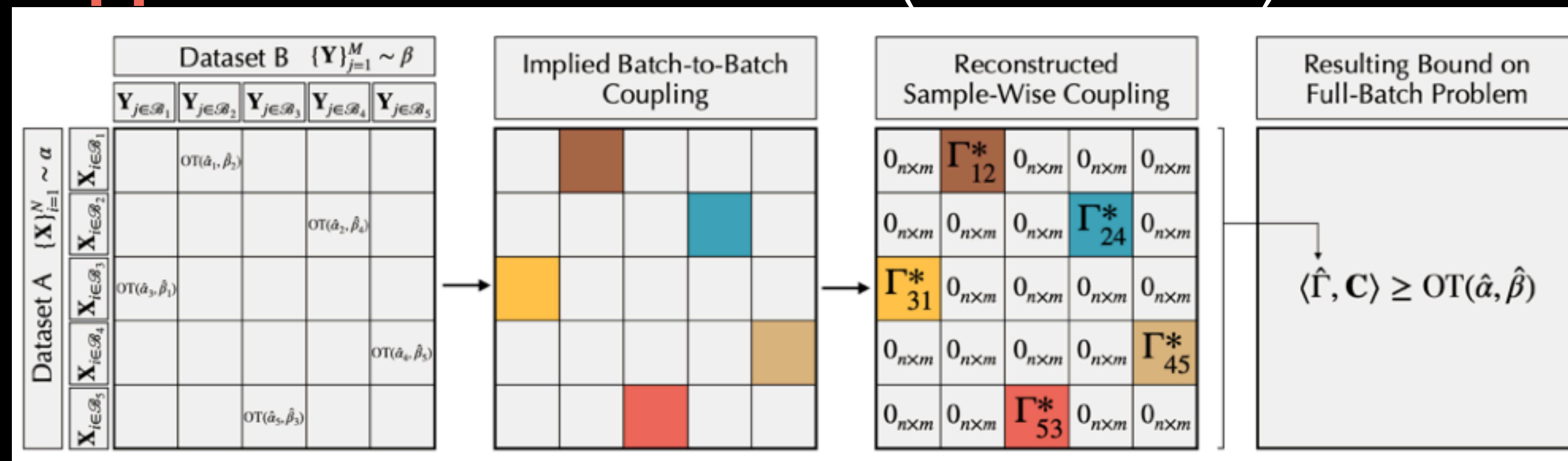
(b) Greedy batch matching



(c) BHOT: optimal batch matching

Mini-Batch Estimation

- Mini-batching gives us **useful gradients** to train models
- But is the OT distance computed by mini-batching + averaging ‘correct’?
- Almost: **it's an upper bound** on the exact (full-dataset) OT distance



- Can be tightened via judicious choice of batch-pairing (AM et al. 2022)



Application: Generative Modeling

Learning Generative Models with Sinkhorn Divergences

Aude Genevay
CEREMADE,
Université Paris-Dauphine

Gabriel Peyré
CNRS and DMA,
École Normale Supérieure

Marco Cuturi
ENSAE CREST
Université Paris-Saclay